Module 08 • Empirical Research

Experimental Evaluation & Academic Benchmarks

IRRIGATEWISE Hydrogel-Based Smart Irrigation System Specification

Experimental Evaluation & Academic Benchmarks

This document records the design, execution, quantitative results, and findings of **six controlled experiments** evaluating the IRRIGATEWISE intelligent irrigation architecture.

Experiment 1: Rule-Based SMD vs. ML Predictive Water Consumption

Objective:

Quantify the water conservation delta between conventional Rule-Based Soil Moisture Deficit (SMD) irrigation and the autonomous ML Predictive Irrigation engine over a simulated 30-day crop growth phase.

Methodology:

- **Plot Area:** 1,000 m^2 loam soil cultivated with Sugarcane (Z_r = 0.5 m, Field Capacity θ_{FC} = 62%, Target $\theta_{optimal}$ = 65%).
- **Trial A (Control - Rule-Based SMD):** Irrigated whenever instantaneous θ < 45%. Volume applied restored soil to 65%.
- **Trial B (ML Predictive + Hydrogel):** Irrigated when forecasted θ_{3h} < 46%. Applied volume incorporated hydrogel retention buffer factor (φ = 0.28) and 24h precipitation awareness.

Quantitative Results:

Metric	Trial A (Rule-Based SMD)	Trial B (ML Predictive + Hydrogel)	Absolute Delta	Percentage Reduction
Total Irrigation Events	21 cycles	15 cycles	-6 cycles	**-28.6%**
Total Water Applied	115,500 Liters	79,800 Liters	-35,700 Liters	**-30.9%**
Mean Crop Moisture	51.2%	53.8%	+2.6%	+5.1% Stability
Hours Below Wilting Point	14.5 hours	1.0 hour	-13.5 hours	**-93.1% Stress**

Key Finding:

ML predictive scheduling with hydrogel buffering saved **30.9% of total irrigation water** while simultaneously eliminating vegetative moisture stress by triggering replenishment 3 hours ahead of critical depletion.

Experiment 2: Soil Moisture Retention Kinetics (Hydrogel vs. Control)

Objective:

Evaluate the continuous drying kinetics of soil amended with 50 g/m² hydrogel versus unamended control soil during a 48-hour continuous desiccation period with zero irrigation.

Methodology:

- Both soils saturated to field capacity (62%) at t = 0.
- Ambient environment subjected to diurnal thermal stress (T_max = 34°C, RH_min = 45%).
- Moisture sampled hourly for 48 hours.

Quantitative Results:

Metric	Control Soil	Hydrogel Amended Soil (50 g/m²)	Improvement
Moisture at 24 Hours	43.2%	**52.8%**	+9.6% higher
Moisture at 48 Hours	32.6% (Near wilting)	**45.1%** (Above stress)	+12.5% higher
Time to Breach 45% Threshold	21 hours	**48 hours**	**+128.5% extension**
Mean Desorption Rate	-0.61% / hour	**-0.35% / hour**	**42.6% slower drying**

Key Finding:

Hydrogel polymers more than double the soil's moisture half-life under peak thermal conditions, extending the required interval between irrigation events from 21 hours to 48 hours.

Experiment 3: Weather-Aware Rainfall Suppression

Objective:

Verify whether upcoming rainfall suppression prevents redundant irrigation events without compromising soil water availability.

Methodology:

- 10 simulated rainfall events ($P \in [6\text{ mm}, 25\text{ mm}]$) scheduled during drying phases.
- **Rule-Based Engine:** Checked only current instantaneous moisture.

- **ML Predictive Engine:** Evaluated 24h quantitative precipitation forecast (P_forecast ≥ 5 mm).

Quantitative Results:

- **Rule-Based Engine:** Fired 8 redundant pump cycles immediately prior to rain events, causing subsequent surface runoff and deep leaching.
- **ML Predictive Engine:** Successfully suppressed all 8 redundant events. Total water saved solely through rainfall awareness: **44,000 Liters**.
- Post-rainfall ground truth moisture reached optimal field capacity (62%) naturally via precipitation in 100% of suppressed cases.

Experiment 4: Real-Time Anomaly Detection & Fault Isolation

Objective:

Measure the precision, recall, and false alarm rate of the multi-condition anomaly screening filter against injected edge failure modes.

Methodology:

Injected 100 synthetic fault packets into the telemetry stream: - 30 Physical Out-of-Bounds (e.g., SM = -5% or 105%). - 30 Sudden Micro-Spikes (>15% jump in 5 min with pump off). - 20 ADC Sensor Freezes (invariant sensor readings for 6 hours). - 20 Pump-Moisture Decoupling events (pump on for 25 min with zero moisture increase).

Quantitative Results:

Injected Fault Category	Injected Events	Correctly Detected	False Positives	Detection Recall
Out-of-Bounds	30	30	0	**100.0%**
Sensor Spikes	30	29	1	**96.7%**
ADC Frozen Value	20	20	0	**100.0%**
Pump Decoupling	20	19	0	**95.0%**
Overall Suite	**100**	**98**	**1**	**98.0%**

Experiment 5: Multi-Horizon Forecasting Error Degradation

Objective:

Quantify error expansion across expanding operational forecasting horizons (1h, 3h, 6h) to determine the boundaries of reliable predictive decision-making.

Quantitative Results:

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Horizon (h) MAE (%) RMSE (%) R² Score Decision Safety
-------------------------------------------------------------
t + 1h  0.751%  1.258%  0.9365  Exceptional (High Certainty)
t + 3h  1.442%  2.915%  0.6585  Optimal Decision Trigger
t + 6h  1.987%  3.774%  0.4633  Strategic Trend Only
```

Finding:

The 3h operational window represents the optimal trade-off between foresight and predictive variance.

Experiment 6: Closed-Loop Feedback & Residual Drift Tracking

Objective:

Verify that the `prediction_feedback` feedback service captures actual ground truth discrepancies and enables continuous auditing of model health.

Methodology:

- Executed 50 consecutive predictions followed by logged ground truth observations at t + 1h.
- Analyzed residuals: $e_t = \theta_{actual} - \hat{\theta}_{pred}$.

Quantitative Results:

- Mean Error (Residual Bias): +0.042% (Virtually unbiased).
- Absolute Error Distribution: 84% of errors < 1.0%, 98% of errors < 2.5%.
- Closed-loop audit trail verified via `GET /api/v1/irrigation/feedback` with zero database transaction collisions.